

Chapter 42 — Bladder Anomalies in Children

- ICD-10-CM (International Classification of Diseases, Tenth Revision, Clinical Modification) classifies birth defects as **structural or functional**; bladder anomalies span both.
- **Structural**: exstrophy, duplication, agenesis, urachal anomalies, diverticular disease. **Functional**: usually **secondary** — bladder outlet obstruction, neurogenic dysfunction. **Both**: the bladder of **prune-belly syndrome (PBS)**.
- Focus is the **less common congenital anomalies** and **rare acquired paediatric conditions**; embryology and prenatal imaging matter because **many congenital anomalies are now first detected prenatally**.

CONGENITAL BLADDER ANOMALIES

Embryology of the Urinary Bladder

- **Weeks 4–6**: the **urorectal septum** divides the endodermal **cloaca** → ventral **urogenital sinus** + dorsal **rectum**.
- **Cranial urogenital sinus** (continuous with the allantois) → **bladder and pelvic urethra**. **Caudal urogenital sinus** → **phallic urethra (male) / distal vagina (female)**.
- **Male urethra** = pelvic + phallic parts; **entire female urethra** = pelvic part of the urogenital sinus only.
- **Allantois**: extraembryonic outpouching of the yolk sac, joining the cranioventral cloaca (future bladder).
- **Week 10**: definitive nephrons start **urine production** — the **earliest point at which lower urinary tract anomalies can show**.
- **Months 4–5**: bladder **descends into the pelvis**; the allantoic duct does not grow with the embryo, so it **elongates and narrows** → obliterates to a fibrous cord, the **urachus** → **median umbilical ligament** (bladder apex to umbilicus).

Concept — *the urachus is one tube obliterated by descent and elongation; every urachal anomaly is a failure of that single process at a different level.*

Normal and Abnormal Antenatal Sonographic Findings of the Bladder

- **Normal** = **absence of abnormal findings**; criteria for abnormality are broad, and **physiological bladder cycling** must be excluded first.
- **Elliptical anechoic pelvic structure**, bordered **laterally by the umbilical arteries**, **anteriorly by the pubic bones**, **posteriorly by the rectosigmoid**; **wall ≤ 3 mm**, echogenicity like other pelvic structures.
- **Visible in about half at 10 weeks** (onset of urine production), rising to **almost all by 13 weeks**; **transvaginal is superior to transabdominal scanning** (seen in the large majority at 12 weeks).
- **Fetal bladder empties every 15–20 minutes** → if not seen, **a repeat scan in the same sitting is mandatory**.
- **First-trimester diameter ≤ 6 –8 mm**.
- **Fetal sex**: **unreliable before 14 weeks**; base on **visualising the testes, not the phallus**.
- **Amniotic fluid**: until **16 WGA (weeks' gestational age)** mainly **placental transudate**, thereafter **predominantly fetal urine** — so it reflects fetal urine output only **after 16 weeks**.
- **Umbilical cord**: **two arteries, one vein, no fluid-filled urachus**.

Concept — *the fetal bladder is a cycling organ, so one image can neither prove absence nor confirm dilation; timing — emptying every 15–20 minutes, amniotic fluid becoming urine at 16 weeks, sex reliable from 14 weeks — governs interpretation.*

Nondilated or Absent Bladder

- A **non-visualised bladder** is **absent, malformed, or present but not filling** — a normal bladder may not fill because of **renal dysfunction** or **ectopic drainage**; **renal development and amniotic fluid** discriminate.
- **Absent bladder is diagnosed only after a repeat scan 15–20 minutes later.**

Bladder Agenesis

- **Mechanism:** the hindgut is usually normal, so cloacal division is regular → either **atrophy of the cranial urogenital sinus** or **failure to incorporate the mesonephric ducts and ureters into the trigone**.
- **Associations:** neurological, orthopaedic and urogenital — **renal dysplasia or agenesis**; absent **prostate, seminal vesicles, penis, vagina**.
- **Incidence about 1:600,000**; few reported cases, a minority live-born; **significant female majority**.
- **Compatible with life only if the ureters drain ectopically** — into **müllerian structures (female)** or the **rectum (male)**.
- **Confirm with antegrade or retrograde contrast; MRI (magnetic resonance imaging) or CT (computed tomography)** for anatomy before surgery. **Preserve renal function by ureterosigmoidostomy or external stoma.**

Bladder Hypoplasia

- The bladder forms and is visible throughout pregnancy but **never reaches adequate capacity** because of **inadequate filling or storage** in fetal life.
- Some **grow once the cause is corrected**; **later augmentation is often required**.

CAUSES OF BLADDER HYPOPLASIA

Mechanism	Example
Inadequate outlet resistance	Severe epispadias
Separation defect	Urogenital sinus anomalies
Abnormal renal development	Bilateral renal dysplasia or agenesis
Urine bypassing the bladder	Ureteral ectopia

Cloacal and Bladder Exstrophy

- **Bladder template only** → suspected when **regular bladder filling is absent**.
- **Covered cloacal exstrophy:** omphalocele, pelvic diastasis and genital anomalies **with bladder filling**; may become exstrophic near birth.
- **Many cases still missed prenatally**, though detection is improving; **fetal MRI** adds anatomical detail for counselling.

NONDILATED OR ABSENT FETAL BLADDER COMPARED

	Agenesis	Hypoplasia	Exstrophy
Bladder on ultrasound	Absent (after repeat scan)	Present, small ; never reaches capacity	No filling; template on abdominal wall
Amniotic fluid	Not stated; depends on renal and ectopic drainage	Not stated	Normal throughout
Core defect	Cranial urogenital sinus atrophy or trigone not formed	Inadequate filling or storage	Open bladder plate

	Agenesis	Hypoplasia	Exstrophy
Outcome	Survival only with ectopic ureteric drainage	May grow; augmentation often needed	Reconstructive pathway

Concept — a non-visualised bladder is a filling problem before it is a structural one; the discriminators are the repeat scan, the kidneys and the amniotic fluid, and exstrophy alone combines absent filling with normal amniotic fluid and a visible abdominal-wall template.

Dilated Bladder and Congenital Megacystis

Megacystis

- Originally attributed to **bladder neck obstruction** causing massive bilateral **VUR (vesicoureteral reflux)** and a thin wall — but **bladder neck surgery did not change outcome**.
- Re-examination showed **normal urethras and complete emptying on VCUG (voiding cystourethrography)**: reflux is the cause, not the consequence, of dilation, through continuous **recycling of urine between upper tracts and bladder**.
- Modern usage: **megacystis = any prenatally dilated bladder**, cause-neutral. Like hydronephrosis, a finding, not a diagnosis — harder to judge because the **bladder is normally distended at times**. Rare (about 0.06% of unselected fetuses).
- First trimester (10–14 WGA)**: normal controls all had **sagittal length <10% of CRL (crown–rump length)**, i.e. **≤6 mm**; established megacystis had a ratio **≥13%** and diameter **≥8 mm**.
- Beyond the first trimester** definitions are mostly **subjective**; later studies focus on a disease such as **PUV (posterior urethral valves)** or **PBS**, where size is one finding among several. **Bladder cycling is the marker of normal function**.
- The formula (15–40 WGA) is **SL (sagittal length, mm) = GA (gestational age, weeks) – 5, 95% CI (confidence interval) ±7**; the two tiers allow **more specific prenatal counselling**.

DEFINITIONS OF FETAL MEGACYSTIS COMPARED

Gestation	Method	Normal	Abnormal
10–14 WGA	Sagittal length; length/CRL	≤6 mm; <10% of CRL	≥8 mm; ≥13% of CRL (other series: >6 mm or ≥7 mm)
After first trimester — functional	Observation of cycling	Empties within 45 min	No emptying in 45 min (commonest definition)
15–40 WGA — formula	SL = GA – 5 (±7)	≤ GA + 2 mm	Enlarged bladder: > GA + 2 mm. Megacystis: > GA + 12 mm

Differential diagnosis.

CAUSES OF MEGACYSTIS: OBSTRUCTIVE VERSUS NON-OBSTRUCTIVE

Obstructive — LUTO (lower urinary tract obstruction)	Non-obstructive
PUV	PBS
Urethral atresia	MMIHS (megacystis–microcolon–intestinal hypoperistalsis syndrome)
Urethral stenosis / anterior urethral valves	Cloacal anomalies (some overlap between groups)

- Meta-analysis of more than 1,000 fetuses**: a diagnosis is reached in the majority; **PUV is commonest**; no cause in over one-third.
- Male:female 8:1**, because the leading cause, PUV, is male-only.

TABLE 42.1 DEFINITIVE DIAGNOSES IN FETUSES WITH MEGACYSTIS

Diagnosis	Mechanism	Pooled proportion
PUV	Obstructive	About 50% (499/998)
Urethral atresia/stenosis	Obstructive	7% (74/998)
PBS	Non-obstructive	4%
MMIHS	Non-obstructive	1%
Cloacal abnormality	Non-obstructive	0.7%
Undefined	—	36.5%

From Taghavi K, Sharpe C, Stringer MD: Fetal megacystis: a systematic review. J Pediatr Urol 13:7–15, 2017 (16 pooled series; obstructive diagnoses over 998 because one series did not specify them, others over 1,067).

- **What it is for:** ranking causes of fetal megacystis. **Pattern:** obstructive causes dominate; the non-obstructive syndromes are individually rare; **"undefined" is the second-largest group.**
- **Trap:** megacystis ≠ PUV — **over one-third never get a diagnosis;** a female fetus with megacystis cannot have PUV → think **MMIHS, cloacal anomaly, urethral atresia;** MMIHS matters far beyond its 1% because of morbidity.

Aneuploidy.

- About **one-fifth of fetuses with megacystis are aneuploid**, mostly **trisomy 13 or 18** — more than maternal age and nuchal translucency predict.
- Outcome **uniformly poor:** most terminated, the rest did not survive to term; high termination **limits urological outcome data.**

Resolution by size and timing (euploid fetuses).

- **First-trimester longitudinal diameter 7–15 mm:** the **large majority resolve** spontaneously; the remainder have obstructive uropathy. **>15 mm: none resolve — all obstructive uropathy.**
- In a national multicentre fetal-medicine database, overall resolution was a minority, but **early presentation resolved about three times as often as late;** **early with ≤12 mm** was a sensitive predictor of resolution (most resolved). Limitation: population selected by high termination; genetics not captured.

EARLY VERSUS LATE MEGACYSTIS

Early (<18 WGA)	Late (>18 WGA)
Resolves more often; ≤12 mm predicts resolution	Resolves less often
Resolved before 23 WGA → no postnatal urological surgery	Resolved after 23 WGA → most still need postnatal intervention despite prenatal resolution

- **Persistent megacystis with oligohydramnios → LUTO.**

Fig. 42.1 — Fetal megacystis at 18 and 38 weeks (Pellegrino et al., 2017): a series showing **prenatally detected megacystis is not always an adverse prognostic factor.**

Megacystis Microcolon Intestinal Hypoperistalsis Syndrome

- About 1% of megacystis but important for its **morbidity.** **Prenatally, megacystis is often the only finding;** gastrointestinal (GI) findings are rarely seen, but **amniotic fluid digestive enzymes are abnormal in most** if sampled.
- **Visceral smooth-muscle dysfunction** of GI and urinary tracts; **ACTG2 mutations** (actin smooth-muscle filament).
- **Survival:** historically **about 20%**, improving to **about 55%** in cases from 2004–2011.

- **Postnatal:** abdominal distension, bilious vomiting, failure to pass meconium — **bowel obstruction with an enlarged bladder = evaluate as MMIHS**. Exclude Hirschsprung disease by colonic biopsy showing ganglion cells; contrast shows a **hypomotile small bowel and a small colon**.
- **Enteric care:** ileostomy, gastrostomy, TPN (total parenteral nutrition); TPN complications — **bloodstream infection and liver dysfunction** — are the leading causes of death; small-intestinal transplantation is promising.
- **Urological care:** morbidity mostly secondary to enteric disease → short-term CIC (clean intermittent catheterisation) or vesicostomy for retention → once enterically stable, urodynamics → **most need lifelong CIC** and lifelong urological follow-up, as in PUV or neurogenic bladder. Normal micturition in older patients is reported but uncommon.

LUTO VERSUS MMIHS

	LUTO (e.g. PUV)	MMIHS
Amniotic fluid	Oligohydramnios	Normal or increased
Sex	Male (PUV)	Female predominance (about 70%)
Mechanism	Anatomical outlet obstruction	Visceral myopathy (ACTG2)
VUR	Common	Uncommon → early contrast cystography of limited value
Bladder on urodynamics	Obstructed, high pressure	Acontractile

Concept — megacystis is a sign whose meaning is set by size, gestation at detection and timing of resolution, with karyotype and amniotic fluid as the overriding qualifiers; **small, early, early-resolving is benign** — **large, late or persistent signals obstruction**, and normal fluid with megacystis in a female points away from obstruction towards MMIHS.

Bladder Duplication

- **Complete or incomplete; sagittal or coronal. Complete sagittal duplication is commonest.**
- **Rarely one bladder lacks a urethra** → complete obstruction → **ipsilateral renal dysplasia**. Continence varies.
- **External genital duplication in up to 90%; lower GI duplication in up to 42%.** Duplicated vagina ± separate unicornuate uterus; each duplicated penis has its own urethra. VUR, renal ectopia and dysplasia are common.
- **Non-urological anomalies are more frequent in sagittal than coronal duplication:** GI malformations, spinal duplication, spina bifida, urogenital–GI fistulae.
- **Exstrophy variant:** exstrophic bladder and urethra **plus** a normal closed intra-abdominal bladder.
- **Presentation:** often neonatal (GI or genital anomalies); many later with **recurrent infection or incontinence**.
- **Work-up:** karyotype, ultrasound, videourodynamics, genitogram, cross-sectional imaging; VCUG and renal nuclear scans for VUR and function. **Final plan often deferred to endoscopy and surgical exploration.**

INCOMPLETE VERSUS COMPLETE DUPLICATION

Incomplete	Complete
Halves communicate ; usually one urethra	Two separate bladders , each with normal mucosa and full-thickness muscle, divided by a peritoneal fold ; each usually with its own ureter, urethra and meatus ; sizes may differ
May need no surgery if both halves drain through the common urethra	Usually combined into one bladder

Treatment sequence: preserve renal function and prevent infection by relieving obstruction → later, **continence and genital reconstruction: both sphincters competent → join distal urethras; one incompetent → close that bladder neck and excise its urethra**; duplicated vaginas joined in the midline with **vulvoplasty**. Asymptomatic patients may be left uncorrected (successful pregnancies reported). Individualise, in **centres experienced in complex urogenital reconstruction**.

Concept — sphincter competence, not bladder anatomy, decides the reconstruction; whatever the configuration, kidneys and infection come first, continence and genitalia later.

Congenital Diverticula

- **Mucosa herniates between detrusor fibre defects; neck width = size of the muscle defect.**
- Low incidence (about 1.7% of symptomatic children imaged); **true incidence higher** because asymptomatic diverticula go undiscovered.
- Detectable prenatally, but mostly found postnatally during work-up for **infection, haematuria, incontinence or obstruction**.
- Ultrasound shows large ones; **VCUG is the gold standard** (shows **VUR** and **post-void retention**); nuclear scan for function and obstruction if hydroureteronephrosis.
- **Dynamic — may not appear on every study**; repeat imaging if suspicion persists.

Paraureteral Congenital Diverticula

- **By far the commonest** congenital diverticula; **Hutch** divided them into **primary and secondary** (secondary are acquired — see Acquired Diverticula).
- **Primary: herniation through the ureteral hiatus, between the intravesical ureter and the roof of the hiatus**, most likely from a **congenitally deficient bladder wall**. An isolated **Waldeyer sheath** defect is unlikely because they are usually unilateral.
- **Small and asymptomatic → conservative. With VUR → many recommend excision.**
- **Females resolve (diverticulum with VUR) spontaneously more often than males** — attributed to **higher male voiding pressures** even without obstruction.
- **A primary paraureteral diverticulum does not preclude endoscopic dextranomer/hyaluronic acid (Deflux) injection for VUR.**

Fig. 42.3 — Large primary paraureteral diverticulum on contrast study and ultrasound.

Posterolateral and Multiple Congenital Diverticula

- Associated with **connective tissue disorders — Ehlers–Danlos, Williams, Menkes** — occasionally non-syndromic.
- Can become **massive, often larger than the bladder** (Fig. 42.5: isolated posterolateral diverticulum in a newborn, larger than the bladder).
- Histology: an **attenuated, lax muscle wall** rather than classic herniation through a defect, **without obstruction**.
- Presents with **recurrent UTI (urinary tract infection)** from stasis (Fig. 42.4: post-void residual in the diverticulum) and **stones**.
- **Asymptomatic → close observation. Symptomatic → resection** (open or laparoscopic) eliminates UTI and often improves voiding.
- **Recurrence and new diverticula are common in syndromic patients** → weigh against quality of life; **vesicostomy or suprapubic drainage** may be preferred. **Postoperative urodynamics** to detect obstruction or high-pressure voiding that would hasten recurrence.

BLADDER DIVERTICULA IN CHILDREN COMPARED

	Primary paraureteral	Secondary paraureteral (acquired)	Posterolateral / multiple (syndromic)
Bladder wall	Smooth	Trabeculated	Attenuated, lax
Number	Single, isolated	One of many	Multiple; can be giant
On imaging	Intermittent	Always present	May be dynamic
Cause	Focal congenital wall deficiency	Infravesical obstruction or neurogenic bladder; also infection, previous surgery	Connective tissue disorder
Obstruction	None	Present — the cause	None
Management	Observe if small; excise if VUR ; Deflux possible	Treat obstruction first ; excise ± reimplant if symptomatic	Observe if asymptomatic; resect for UTI or stones; high recurrence

Concept — diverticular behaviour follows the wall and the pressure: a focal defect at normal pressure gives a single intermittent primary diverticulum, generalised laxity gives giant recurrent syndromic ones, and raised pressure gives multiple secondary diverticula — which is why the outlet is treated before the diverticulum.

KEY POINTS: CONGENITAL BLADDER ANOMALIES

- **Persistent megacystis with oligohydramnios → LUTO**; megacystis resolving before delivery may have no postnatal sequelae.
- **MMIHS** may show only megacystis prenatally.
- **Nondilated anomalies** such as exstrophy are the most severe congenital urological malformations; amniotic fluid is usually normal in exstrophy.
- **Bladder agenesis** is compatible with life only with ectopic ureteric drainage.
- **Connective tissue disorders → multiple diverticula**, which may need treatment for **stasis-related infection or stones**.

Urachal Anomalies

- **Symptomatic remnant → complete excision with a bladder cuff**, open or laparoscopic; simple, low morbidity.
- **The asymptomatic incidental remnant is the dilemma**: urachal adenocarcinoma is unreported in children, but in adults carries **high mortality and much more complex surgery**.

THE INCIDENTAL ASYMPTOMATIC URACHAL REMNANT: TWO POSITIONS

Excise all	Excise only if symptomatic
To prevent progression to adult malignancy	Quantified risk of progression is low
Earlier position	Increasingly adopted ; discuss with the family either way

Patent Urachus

- **Non-descent of the bladder** or, more commonly, **failure of the epithelial canal to obliterate**.
- **Fetal obstruction theory is doubtful**: patency is often absent in severely obstructed fetal bladders, and only a minority (up to 14%) have confirmed in-utero obstruction → obliteration appears **independent of distension**.
- **Retubularisation** of an obliterated urachus, rather than primary patency, may explain umbilical urine drainage — supported by umbilical urinary fistulae in obstruction acquired later in life.

- **Neonatal umbilical drainage, enlarged or oedematous umbilicus, delayed cord-stump healing.**
Organisms: *Staphylococcus aureus*, *Escherichia coli*, *Enterococcus*, *Citrobacter*, rarely *Proteus*.
- **Diagnosis: fluid-filled canal on longitudinal ultrasound**, or contrast on **retrograde fistulogram or VCUG** (Fig. 42.7: contrast pooling in the umbilicus); MRI depends on bladder filling. **Differentiate from a patent omphalomesenteric duct** (coexistence rare).
- **Infected or abscessed: drain under antibiotics → complete excision with a bladder cuff** once settled; **remove all anomalous tissue** to prevent recurrence, stones and adenocarcinoma.

Open excision: balloon catheter distends bladder → **transverse or midline infraumbilical incision** (infants: **small transverse subumbilical**, as the dome is still high) → feeding tube into the urachus for identification → circumscribe urachus at the umbilicus, **preserving the umbilicus** → **longitudinal rectus fascia incision**, muscles spread, dome identified → urachus isolated on a vessel loop → **extraperitoneal dissection to the umbilicus**, umbilical portion delivered and removed → **bladder cuff including the insertion** excised with electrocautery → **two-layer watertight closure** → catheter overnight.

- **Laparoscopic excision is feasible even under 6 months** and safe at all ages; series report no complications or recurrences and short operating times.
- **Ports:** three-port — camera **midline between umbilicus and xiphoid**, working ports in each upper quadrant; or camera in the **left lower abdomen** with **2-mm ports left lower and upper** for a better view of the umbilical end. Robotic and single-port approaches are described.

OPEN VERSUS LAPAROSCOPIC URACHAL EXCISION

Open	Laparoscopic
Extraperitoneal; small incision, umbilicus preserved	Intraperitoneal; good visualisation of the urachal course and dome
No peritoneal contamination	Risk of spilling infected or malignant material and intraperitoneal urine leak

Urachal Sinus

- Obliterated at the bladder, **open at the umbilicus**; the caudal urachus is filled with desquamated epithelium.
- Mimic: **persistent omphalomesenteric duct** (a **Meckel diverticulum connected to the umbilicus**) — the **sinugram cannot separate them**, since neither shows a bladder or bowel connection; **both need complete excision**; **omphalomesenteric remnants may show gastric or small-bowel mucosa** on histology.

Urachal Cyst

- Usually **distal**; **commoner in adults than children**; contents of desquamated epithelium; ***S. aureus* commonest** when infected.
- Infected: umbilical abscess, bladder infection, lower abdominal pain, voiding symptoms, painful mass. Ultrasound shows the **cyst between anterior abdominal wall and peritoneum**; **CT or MRI** if extensive.
- **Unrecognised → perforation into bladder or peritoneum** → peritonitis, enteric fistula. **Drain → complete excision.**

Urachal Diverticulum

- Usually **asymptomatic**, incidental; **wide mouth drains well** even if enlarged by obstruction. **Intervene only if a narrow neck causes stones or UTI.**

URACHAL ANOMALIES COMPARED (FIG. 42.6)

	Patent urachus	Urachal sinus	Urachal cyst	Vesicourachal diverticulum
Left open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Communicates with	Bladder + umbilicus	Umbilicus only	Neither (intermittent drainage either way)	Bladder only
Presentation	Neonatal umbilical urine	Continuous draining sinus	Infection ; commoner in adults	Incidental
Diagnosis	Ultrasound; fistulogram or VCUG	Sinugram (no bladder connection)	Ultrasound ; CT/MRI if complex	Incidental imaging
Treatment	Excise with cuff (drain first if infected)	Complete excision	Drain → excise	Only if symptomatic

KEY POINTS: URACHAL ANOMALIES

- Usually detected postnatally because of umbilical drainage.
- Infected remnant → antibiotics, then surgical excision.
- May be confused with a persistent omphalomesenteric duct preoperatively.
- Discuss excision of an asymptomatic incidental remnant with the family, given potential adult malignant transformation.

Concept — every urachal anomaly is the same tube obliterated to a different level, so the level of residual patency predicts the communication, the symptom and the diagnostic test; treatment converges on **complete excision with a bladder cuff**, the only live debate being the asymptomatic incidental remnant.

Concept — congenital bladder anomalies are read through function before form — whether the bladder fills, empties and holds its wall — and both the prenatal picture and the postnatal treatment follow from those three questions.

ACQUIRED BLADDER ANOMALIES IN CHILDREN

Noninflammatory Acquired Conditions

Acquired Diverticula

- Result of **infravesical obstruction**, which is treated first; once outlet resistance is normal, **remodelling** may make diverticulectomy unnecessary.
- **Symptomatic** → excise, with **ipsilateral ureteric reimplantation** if the ureter lies near or within the diverticulum; intravesical or extravesical, open or laparoscopic/robotic.
- **Secondary paraureteral (Hutch)**: from **infravesical obstruction or neurogenic bladder** — raised pressure forces mucosa between muscle fibres, usually **one of many "pop-off" mechanisms**; also from **muscle weakened by infection** or a defect after bladder surgery.

Progression (both paraureteral types): enlarging diverticulum damages the **Waldeyer sheath** → intravesical ureter pulled out of its anchored position → **ureterovesical junction dysfunction** → eventually **ureteric obstruction** → renal dysplasia reported.

- **Vicious circle**: a large paraureteral or posterolateral diverticulum **compresses the bladder neck or posterior urethra** → outlet obstruction → diverticulum fills and expands → **complete retention**; seen with the congenital diverticular syndromes, leading to surgery.

Bladder Haemangiomas

- **Benign vascular tumours**, mostly with **Klippel-Trénaunay or Sturge-Weber syndrome**; usually **solitary**, <3 cm, can be multifocal; present with **gross haematuria** ± irritative symptoms.

Fibroepithelial Polyps

- **Rare in the bladder**; haematuria, irritative symptoms; outlet or ureteric obstruction reported. **Second commonest paediatric bladder tumour after rhabdomyosarcoma**.

Inflammatory Acquired Conditions

Nephrogenic Adenoma and Cystitis Cystica

- **Nephrogenic adenoma**: rare benign tumour, mostly adult; in children a **reaction to infection, stones, trauma or surgery**. Presents with **haematuria or irritative symptoms**; diagnosis by **cystoscopy and biopsy**. **No malignant change reported in children**; long-term resolution reported with **ibuprofen and trimethoprim-sulfamethoxazole** after resection.
- **Malignant potential**: in a young adult with post-traumatic neurogenic bladder, repeatedly resected nephrogenic adenoma became adenocarcinoma → radical cystectomy.
- In biopsy series **most had bladder augmentation with recurrent stones and infection**; immunohistochemistry indicates **origin from distal renal tubular cells**.
- **Cystitis cystica**: fluid-filled cystic masses from **chronic inflammation**. **Cystitis glandularis**: rarer in children; **metaplasia to mucus-secreting epithelium**; **premalignant potential poorly supported**.

Eosinophilic Cystitis

- Sporadic case reports; cause unclear — **immunological disease and allergy** suggested. Histology: **eosinophils throughout all layers** of the wall.
- Dysuria, haematuria, suprapubic pain, **retention** (Fig. 42.8: mass on ultrasound in a teenage boy with severe dysuria and gross haematuria; biopsy showed eosinophilic infiltrate, no malignancy).
- **Transurethral resection with corticosteroids, antihistamines or antibiotics** most successful across ages; a small series treated with all three combined succeeded in every case.

Haemorrhagic Cystitis

- **Almost always related to treatment of childhood malignancy**, especially haematological.
- **Commonest risk factor**: cyclophosphamide or ifosfamide, **exacerbated by BK virus, CMV (cytomegalovirus) and adenovirus** in the immunocompromised child; **pelvic radiotherapy** also causes it.
- **Irrigation is difficult and traumatic** in children (small catheters only) → better results with an **early-detection protocol, catheter avoidance and oral pentosan polysulfate sodium**.

PAEDIATRIC BLADDER LESIONS PRESENTING WITH HAEMATURIA OR A MASS

Lesion	Sex	Association / trigger	Diagnosis	Treatment	Key point
Haemangioma	—	Klippel-Trénaunay, Sturge-Weber	Cystoscopy, biopsy	Small: fulguration or Nd:YAG (neodymium:yttrium-aluminium-garnet) laser. Large or rapid bleeding: partial cystectomy	Solitary, <3 cm
Fibroepithelial polyp	—	—	Cystoscopy	Endoscopic resection — curative	Second to rhabdomyosarcoma

Lesion	Sex	Association / trigger	Diagnosis	Treatment	Key point
Nephrogenic adenoma	Female 5:1	Infection, stones, trauma, surgery; augmentation	Cystoscopy + biopsy	Transurethral fulguration or resection	80% recur, latency about 4 years → surveillance cystoscopy
Cystitis cystica	Almost exclusively female	Chronic inflammation	Cystoscopy	Long-term antibiotic prophylaxis	Mimics tumour
Eosinophilic cystitis	Male	Allergy / immunological (proposed)	Biopsy mandatory — no typical appearance	TUR + steroids, antihistamines, antibiotics; neonates and young children: may self-limit → observe	Eosinophils in all layers
Haemorrhagic cystitis	—	Cyclophosphamide / ifosfamide; viruses; radiotherapy	Clinical	Prevent with mesna (sodium 2-mercaptoethane sulfonate); oral pentosan	Irrigation traumatic in children

KEY POINTS: ACQUIRED BLADDER ANOMALIES IN CHILDREN

- **Acquired diverticula** can cause **obstruction** or **stasis** with recurrent UTI.
- **Routine cystoscopy** after resection of nephrogenic adenoma — **80% recurrence** in children.
- **Cystitis cystica** can **mimic tumour**; best treated with **long-term antibiotic prophylaxis**.
- **Eosinophilic cystitis** has **no typical appearance** and **needs biopsy**.
- **Cyclophosphamide** causes **haemorrhagic cystitis**; **mesna** is given simultaneously for prevention, and oral pentosan polysulfate shows promise.

Concept — acquired lesions divide by what drives them — *pressure (diverticula: treat the outlet first), vascular or stromal growth (haemangioma, polyp: local ablation or resection), and chronic irritation or toxicity (inflammatory lesions: remove the stimulus, biopsy because appearance misleads, and survey for recurrence).*